

What Is A Numerical Method ?

A **numerical method** is a set of arithmetic operations designed to solve a mathematical problem.

Arithmetic Operations:

- Addition
- Subtraction
- Multiplication
- Division

Mathematical problems arisen in the field of engineering can be formulated mainly in the form of an equation or a system of equations which are **differential equations** or **algebraic equations**.

A numerical method can be an one-time calculation or an iterative calculation.

One-Time Calculation:

The method is carried out only once.

Iterative Calculation:

The method is carried out repeatedly until a stopping criterion is met. It may be

- 1) a number of iterations or
- 2) a specified tolerance.

Mostly, the method which is iterative and stopped by a specified **tolerance** is employed in process of problem solving.

Why Do We Use Numerical Methods ?

